

Sampling Methods in Data Science

Lecture Notes

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Learning Objectives

By the end of this lecture, you should be able to:

1. Understand the fundamental difference between population and sample
2. Explain why sampling is necessary in data science
3. Describe and implement three main sampling techniques
4. Calculate sample statistics and make population inferences
5. Apply sampling methods to real-world data science problems

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📅 Week 5: Sampling Methods

Topics Covered:

- Introduction to Sampling: Population vs Sample
- Why Sampling is Essential in Data Science
- Random Sampling Technique
- Systematic Sampling Technique
- Stratified Sampling Technique
- Mathematical Foundations and Applications

1 Introduction: Population vs Sample

💡 Theoretical Foundations

Key Definitions:

Population Size: N

Sample Size: n

Population Mean: $\mu = \frac{1}{N} \sum_{i=1}^N X_i$

Sample Mean: $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$

Population Variance: $\sigma^2 = \frac{1}{N} \sum_{i=1}^N (X_i - \mu)^2$

Sample Variance: $s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$

1.1 The Fundamental Problem

In data science, we often face the challenge of studying large populations. The **population** refers to the complete set of items or individuals we're interested in studying.

Step-by-Step Example

Example: Vegetarian Survey in a Neighborhood

- **Population:** 100 families in your neighborhood
- **Question:** Are most families vegetarian or non-vegetarian?
- **Direct Approach:** Visit all 100 families
- **Time Required:** 5 minutes per family $\times$ 100 families = 500 minutes (8.3 hours)
- **Feasibility:** Practical and achievable

Mathematical Representation:

$$\text{Let } X_i = \begin{cases} 1 & \text{if family } i \text{ is vegetarian} \\ 0 & \text{if family } i \text{ is non-vegetarian} \end{cases}$$

$$\text{Population Proportion: } P = \frac{1}{100} \sum_{i=1}^{100} X_i$$

1.2 The Scaling Problem

Step-by-Step Example

Example: Vegetarian Survey for a City

- **Population:** 5,000,000 residents in a city
- **Question:** Are most residents vegetarian or non-vegetarian?
- **Direct Approach:** Interview all 5,000,000 residents
- **Time Required:** 5 minutes $\times$ 5,000,000 = 25,000,000 minutes

$$= 416,667 \text{ hours}$$

$$= 17,361 \text{ days}$$

$$\approx 47.6 \text{ years (working 24/7)}$$

- **Feasibility:** Impractical and impossible

Sampling Solution:

- Sample 1,000 residents (0.02% of population)
- Time: 5,000 minutes 83 hours
- Statistical validity: Results within $\pm 3\%$ margin of error

2 Why Sampling is Essential

2.1 Mathematical Justification

💡 Theoretical Foundations

Central Limit Theorem (Simplified Version): For large enough sample size n , the sample mean $\bar{x}$ is approximately normally distributed:

$$\bar{x} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

Margin of Error Formula: For a proportion p with sample size n :

$$\text{Margin of Error} = z_{\alpha/2} \times \sqrt{\frac{p(1-p)}{n}}$$

where $z_{\alpha/2}$ is the critical value from standard normal distribution.

📊 Step-by-Step Example

Example: Calculating Sample Accuracy

Suppose we survey 1,000 city residents and find 600 are vegetarian:

$$\text{Sample proportion: } \hat{p} = \frac{600}{1000} = 0.60$$

$$\text{Standard error: } SE = \sqrt{\frac{0.60 \times 0.40}{1000}} = \sqrt{0.00024} \approx 0.0155$$

$$95\% \text{ Margin of error: } 1.96 \times 0.0155 \approx 0.0304$$

$$95\% \text{ Confidence Interval: } 0.60 \pm 0.0304 = (0.5696, 0.6304)$$

Interpretation: We are 95% confident that between 56.96% and 63.04% of the entire city population is vegetarian.

3 Sampling Techniques

3.1 1. Random Sampling ()

💡 Theoretical Foundations

Definition: Every element in the population has an equal probability of being selected.

Mathematical Property:

$$P(\text{Element } i \text{ is selected}) = \frac{n}{N} \quad \forall i \in \{1, 2, \dots, N\}$$

Selection Independence:

$$P(\text{Element } i \text{ AND } j \text{ selected}) = \frac{n}{N} \times \frac{n-1}{N-1}$$

Step-by-Step Example

Example: Random Sampling from Student Database

- Population: 10,000 students in university database
- Required sample: 500 students
- Probability for each student: $\frac{500}{10000} = 0.05 = 5\%$
- Implementation using random number generator

Step-by-Step Procedure:

1. Assign each student a unique ID from 1 to 10,000
2. Generate 500 random integers between 1 and 10,000
3. Select students corresponding to these random numbers
4. If a number repeats, generate another unique number

Advantages:

- Unbiased selection
- Every student has equal chance
- Simple to understand and implement

3.2 2. Systematic Sampling ()

Theoretical Foundations

Definition: Selecting elements at regular intervals from an ordered list.

Mathematical Procedure:

1. Calculate sampling interval: $k = \frac{N}{n}$
2. Randomly select starting point r , where $1 \leq r \leq k$
3. Selected elements: $r, r + k, r + 2k, \dots, r + (n - 1)k$

Step-by-Step Example

Example: Systematic Sampling of Factory Products

- Population: 5,000 products produced daily
- Required sample: 100 products for quality check
- Sampling interval: $k = \frac{5000}{100} = 50$

Random Start Selection:

- Choose random number between 1 and 50, say $r = 17$
- Selected products: 17, 67, 117, 167, ..., 4967

Mathematical Representation:

$$\text{Selected products} = \{17 + 50(i - 1) \mid i = 1, 2, \dots, 100\}$$

Advantages:

- Even spread across entire production
- Simple to implement
- Good for periodic quality checks

Disadvantages:

- May miss patterns if data has periodicity
- Requires ordered list

3.3 3. Stratified Sampling ()

Theoretical Foundations

Definition: Divide population into homogeneous subgroups (strata) and sample from each.

Mathematical Formulas:

- **Population divided into H strata**
- **Strata sizes:** $N_1, N_2, \dots, N_H$ where $\sum_{h=1}^H N_h = N$
- **Sample sizes:** $n_1, n_2, \dots, n_H$ where $\sum_{h=1}^H n_h = n$

Two Allocation Methods:

1. **Proportional Allocation:** $n_h = \frac{N_h}{N} \times n$
2. **Equal Allocation:** $n_h = \frac{n}{H}$ (equal samples from each stratum)

Step-by-Step Example

Example: University Survey by Gender

- Population: 10,000 students (6,000 male, 4,000 female)
- Required sample: 1,000 students

Proportional Allocation:

$$\begin{aligned}\text{Male sample: } n_1 &= \frac{6000}{10000} \times 1000 = 600 \\ \text{Female sample: } n_2 &= \frac{4000}{10000} \times 1000 = 400\end{aligned}$$

Equal Allocation:

$$\begin{aligned}\text{Male sample: } n_1 &= 1000/2 = 500 \\ \text{Female sample: } n_2 &= 1000/2 = 500\end{aligned}$$

Procedure:

1. Create two strata: Male students and Female students
2. Within each stratum, use random sampling
3. Combine samples for final analysis

Advantages:

- Ensures representation from all subgroups
- Reduces sampling error when strata are homogeneous
- Allows separate analysis of each stratum

4 Mathematical Comparison of Sampling Methods

Theoretical Foundations

Variance Formulas for Different Methods:

1. Simple Random Sampling:

$$\text{Var}(\bar{x}) = \frac{\sigma^2}{n} \times \left(1 - \frac{n}{N}\right)$$

2. Stratified Random Sampling:

$$\text{Var}(\bar{x}_{\text{strat}}) = \sum_{h=1}^H \left(\frac{N_h}{N}\right)^2 \frac{\sigma_h^2}{n_h} \left(1 - \frac{n_h}{N_h}\right)$$

3. Systematic Sampling:

$$\text{Var}(\bar{x}_{\text{sys}}) \approx \frac{\sigma^2}{n} \times (1 + (n-1)\rho)$$

where ρ is intra-class correlation

5 Applications in Data Science & Machine Learning

Data Science Applications

A/B Testing for Website Optimization

- **Problem:** Test new website feature
- **Population:** All website visitors (millions)
- **Sampling Approach:** Random sampling to create:
 - Control group (current version)
 - Treatment group (new feature)
- **Sample Size Calculation:**

$$n = \frac{(z_{\alpha/2} + z_{\beta})^2 \times (\sigma_1^2 + \sigma_2^2)}{(\mu_1 - \mu_2)^2}$$

where:

- α : Type I error rate (usually 0.05)
- β : Type II error rate (usually 0.20)
- $\mu_1 - \mu_2$: Minimum detectable effect

Machine Learning: Train-Test Split

- **Problem:** Avoid overfitting in ML models
- **Solution:** Random sampling to split data
 - Training set (70-80%): Model learns patterns
 - Test set (20-30%): Evaluate model performance
- **k-Fold Cross Validation:**
 - Systematic sampling to create k folds
 - Each fold: $\frac{1}{k}$ of data
 - Train on k-1 folds, test on 1 fold
 - Repeat k times, average results
- **Mathematical Representation:** For dataset of size N and k folds:

$$\text{Fold size} = \frac{N}{k}$$

Model trained k times, tested on different $\frac{N}{k}$ samples each time

6 Practice Problems

Practice Problem

Problem 1: Sample Size Calculation

A company wants to estimate average transaction value. From pilot study:

- Standard deviation: \$50
- Desired margin of error: \$5
- Confidence level: 95%
- Population: 50,000 customers

Calculate required sample size.

Solution:

$$\text{Formula: } n = \frac{z^2 \times \sigma^2}{E^2}$$

$$z = 1.96 \quad (\text{for 95\% confidence})$$

$$\sigma = 50$$

$$E = 5$$

$$\begin{aligned} n &= \frac{1.96^2 \times 50^2}{5^2} \\ &= \frac{3.8416 \times 2500}{25} \\ &= \frac{9604}{25} = 384.16 \end{aligned}$$

$$\text{Rounded up: } n = 385$$

Practice Problem

Problem 2: Stratified Sampling Allocation

University has 10,000 students:

- Undergraduates: 6,000
- Graduates: 4,000

Need sample of 1,000 students. Calculate samples using:

- (a) Proportional allocation
- (b) Equal allocation

Solution:

(a) **Proportional Allocation:**

$$\begin{aligned}\text{Undergraduates: } n_1 &= \frac{6000}{10000} \times 1000 = 600 \\ \text{Graduates: } n_2 &= \frac{4000}{10000} \times 1000 = 400\end{aligned}$$

(b) **Equal Allocation:**

$$\begin{aligned}\text{Undergraduates: } n_1 &= 1000/2 = 500 \\ \text{Graduates: } n_2 &= 1000/2 = 500\end{aligned}$$

Practice Problem

Problem 3: Systematic Sampling

Factory produces 5,000 items daily. Quality check needs 100 items. Using systematic sampling with random start 17, list first 5 items to check.

Solution:

$$\text{Sampling interval: } k = \frac{5000}{100} = 50$$

$$\text{Random start: } r = 17$$

Selected items: 17, 67, 117, 167, 217, ..., 4967

First 5 items: 17, 67, 117, 167, 217

Practice Problem

Problem 4: Confidence Interval from Sample

Random sample of 400 voters shows 220 support Candidate A. Construct 95% confidence interval for true proportion.

Solution:

$$\text{Sample proportion: } \hat{p} = \frac{220}{400} = 0.55$$

$$\text{Standard error: } SE = \sqrt{\frac{0.55 \times 0.45}{400}} = \sqrt{0.00061875} \approx 0.02487$$

$$\text{Margin of error: } ME = 1.96 \times 0.02487 \approx 0.04875$$

$$95\% \text{ CI: } 0.55 \pm 0.04875 = (0.50125, 0.59875)$$

Interpretation: We're 95% confident true proportion is between 50.1% and 59.9%

Practice Problem

Problem 5: Comparing Sampling Methods

City has 1 million residents:

- North: 400,000
- South: 300,000
- East: 200,000
- West: 100,000

Need sample of 10,000. Compare sample compositions using:

- (a) Simple random sampling
- (b) Stratified sampling with proportional allocation

Solution:

- (a) **Simple Random Sampling:** Approximately proportional by chance, but not guaranteed

- (b) **Stratified with Proportional Allocation:**

$$\text{North: } (400,000/1,000,000) \times 10,000 = 4,000$$

$$\text{South: } (300,000/1,000,000) \times 10,000 = 3,000$$

$$\text{East: } (200,000/1,000,000) \times 10,000 = 2,000$$

$$\text{West: } (100,000/1,000,000) \times 10,000 = 1,000$$

Advantage of stratification: Guarantees representation from all regions, reducing potential bias.

7 Summary and Key Takeaways

Method	Best For	Mathematical Advantage	Real-world Example
Random	Homogeneous populations	Unbiased, simple analysis	Online survey
Systematic	Ordered lists, production	Even spread, simple	Quality control
Stratified	Heterogeneous subgroups	Reduces variance, ensures representation	Political polls

Table 1: Comparison of Sampling Methods

7.1 Key Formulas Summary

1. Sample Size Calculation:

- For means: $n = \frac{z^2 \sigma^2}{E^2}$
- For proportions: $n = \frac{z^2 p(1-p)}{E^2}$

2. Confidence Interval:

- For means: $\bar{x} \pm z \frac{\sigma}{\sqrt{n}}$
- For proportions: $\hat{p} \pm z \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$

3. Sampling Error:

$$\text{Standard Error} = \frac{\sigma}{\sqrt{n}}, \quad \text{Margin of Error} = z \times \text{Standard Error}$$

7.2 Practical Guidelines

1. **Always define population clearly** before sampling
2. **Choose method based on:**
 - Population characteristics
 - Available resources
 - Required precision
3. **Calculate required sample size** before data collection
4. **Document sampling procedure** for reproducibility
5. **Report margin of error** with all estimates

7.3 Common Pitfalls to Avoid

- **Sampling bias:** Ensure equal probability of selection
- **Under-sampling:** Too small sample $\rightarrow$ wide confidence intervals
- **Over-sampling:** Wastes resources with diminishing returns
- **Ignoring stratification:** Misses opportunity to reduce variance
- **Non-response bias:** Those who don't respond may differ systematically

Book Chapters & Online Resources

Recommended Book Chapters:

1. **Casella & Berger:** "Statistical Inference" - Chapter 5: Properties of a Random Sample
2. **James et al.:** "An Introduction to Statistical Learning" - Chapter 5: Resampling Methods
3. **Wasserman:** "All of Statistics" - Chapter 8: Estimating the CDF and Statistical Functionals

Online Resources:

- [Seeing Theory](#) - Interactive visualizations
- [Khan Academy: Statistics](#)
- [StatTrek: Online Tutorials](#)

Next Week: Central Limit Theorem and Sampling Distributions

Learning Resources

- **Python Libraries:** `scipy.stats`, `statsmodels`, `scikit-learn`
- **R Packages:** `infer`, `boot`, `caret`
- **Online Courses:** Coursera - Data Science Statistics, edX - Probability and Statistics
- **Interactive Tools:** [Sample Size Calculator](#)